



OthelloAI is a concrete application of the knowledge we've gained since the beginning of the year, combined with personal research and exploration. It consists of an implementation of the classic Othello game, integrating various technologies:

- **Python** for audio processing, and communication protocols
- **HTML, CSS, and JavaScript** for the user interface and AI logic.

This project allows users to play Othello online, either solo against an AI or in a multiplayer setup. It showcases our ability to combine different areas of development AI, web design, and real-time communication into a unified, interactive experience.

```

└─ OthelloAI/
  └─ ai
  └─ communication
  └─ res
  └─ static/
    └─ css
    └─ images
    └─ sounds
    └─ js/
      └─ game.js
  └─ templates/
    └─ index.html
  └─ app.py
  └─ audio.py
  └─ Dockerfile
  └─ README.md
  └─ Requirements.txt

```

PROJECT ARCHITECTURE

Backend: Flask server (app.py) to handle the APIs (like audio one).

Frontend: User interface in HTML/CSS/JS (index.html, static/), also handles game logic through game.js.

AI: Advanced algorithms like NegaScout for AI decision-making.

Voice Recognition: audio.py module to interpret voice commands.

Dockerfile: Allows the project to run without worrying about the specific technical setup of each user.

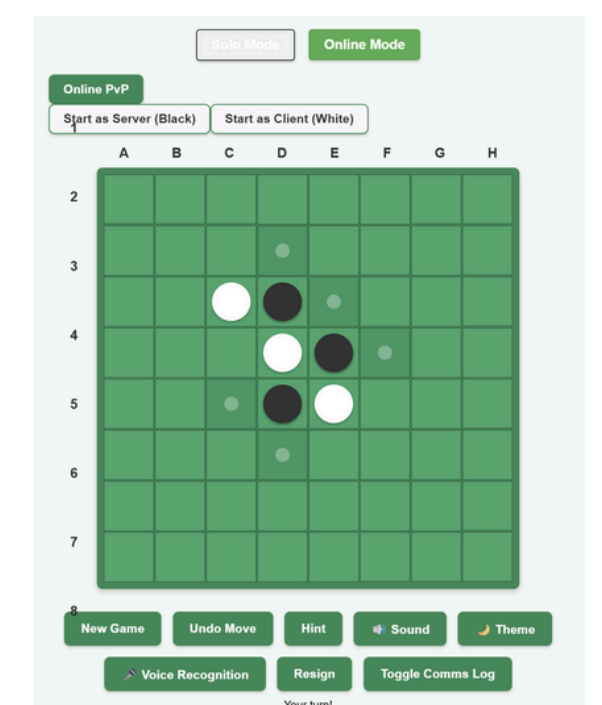
COMMUNICATION

How it works

Two Raspberry Pi devices play Othello over a TCP/IP network. A Flask server manages the game logic and state. The game can be started as either a server or a client. Additionally, a communication log is available.

Message Format: <header>|<body>

- Header: Action type
- Body: Parameters (if any), separated by |



Communication Log

```

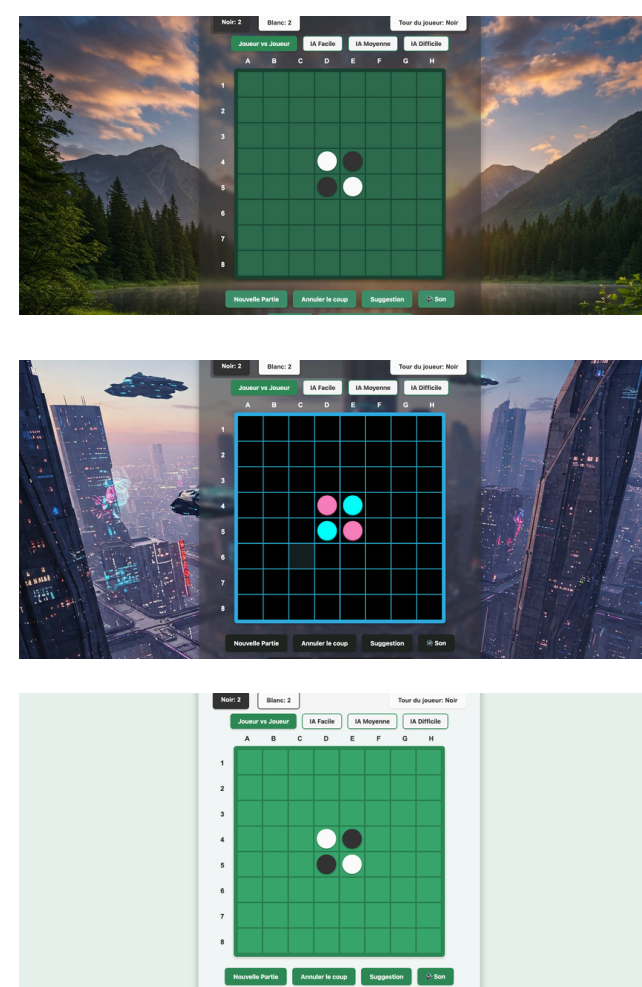
[20:23:55] [INFO] Assigned role: P1 (Black/Server)
[20:23:56] [SUCCESS] Connected to server as server with role P1
[20:23:56] [INFO] Server assigned role: P1
[20:24:00] [SENT] Sending move D3 as P1
[20:24:00] [SENT] Sending: 1|03|B
[20:24:00] [RECEIVED] Received: -1
[20:24:00] [SUCCESS] Received OK acknowledgment (Code -1)
[20:24:00] [SUCCESS] Server responded OK (-1)

```

INTERFACE

A minimalist and modern design centered around user experience

- **Dynamic score** display
- **Game modes** (Player vs Player, Easy/Medium/Hard AI)
- **Control bar:** new game, undo, hints, sound control, theme switching, and voice recognition
- **Smooth animations** and subtle visual effects
- **Theme** system to customize the game's appearance



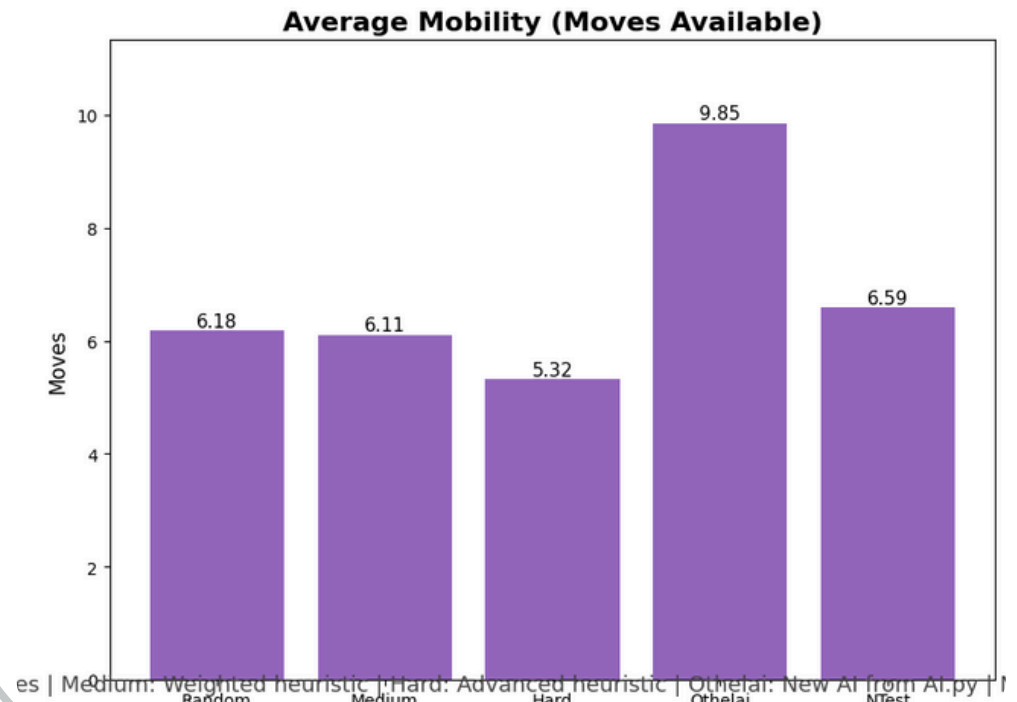
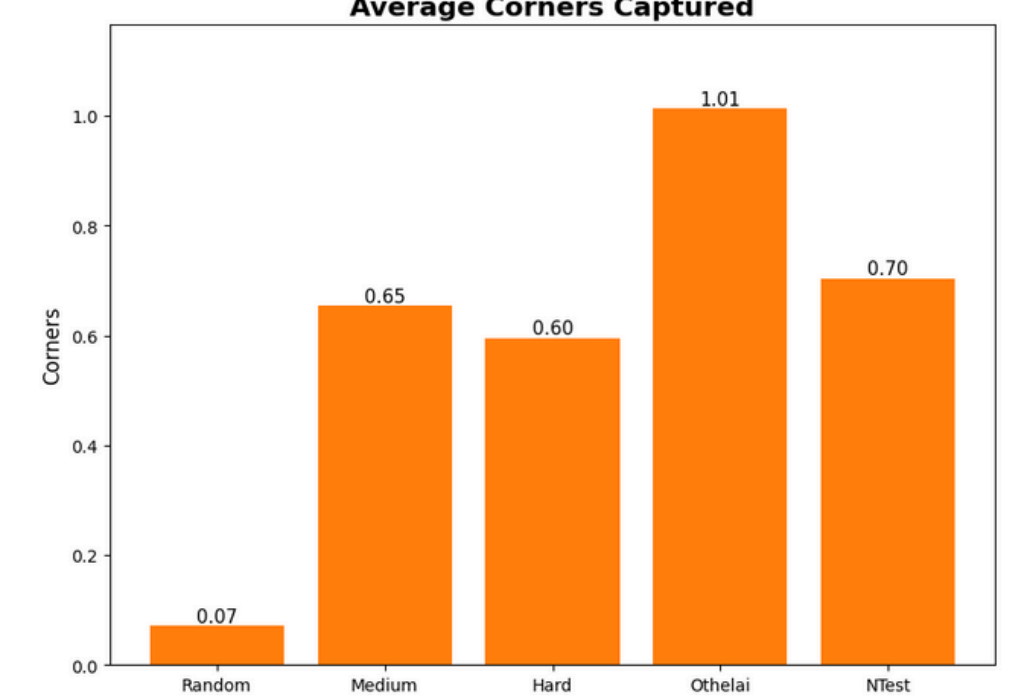
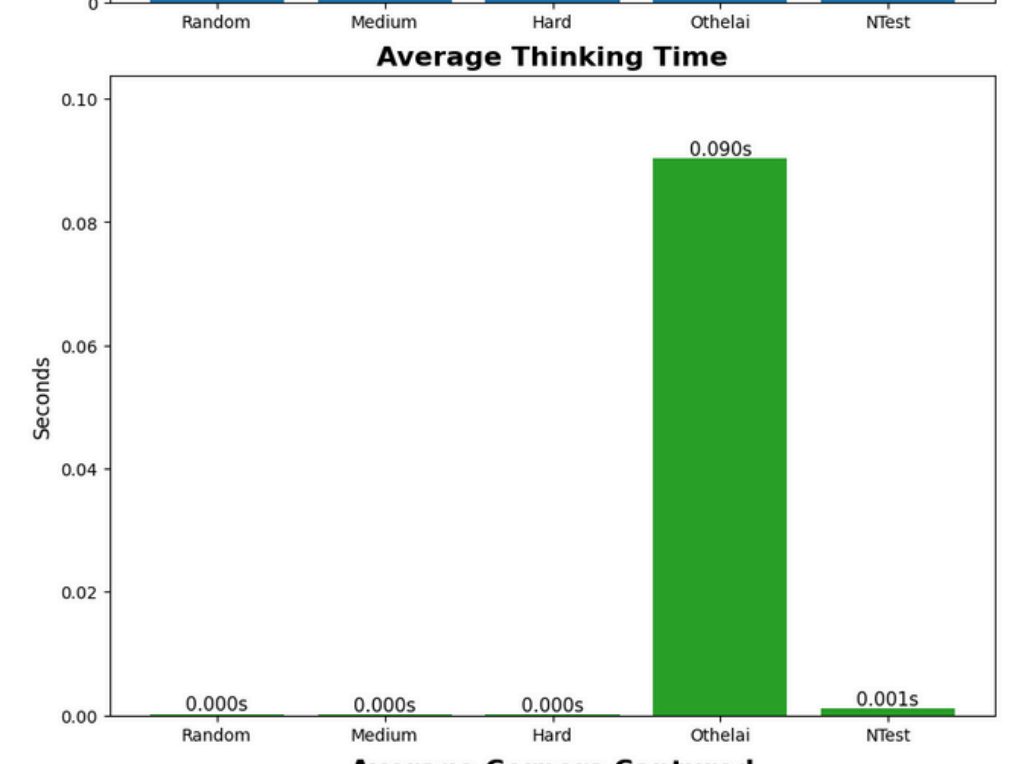
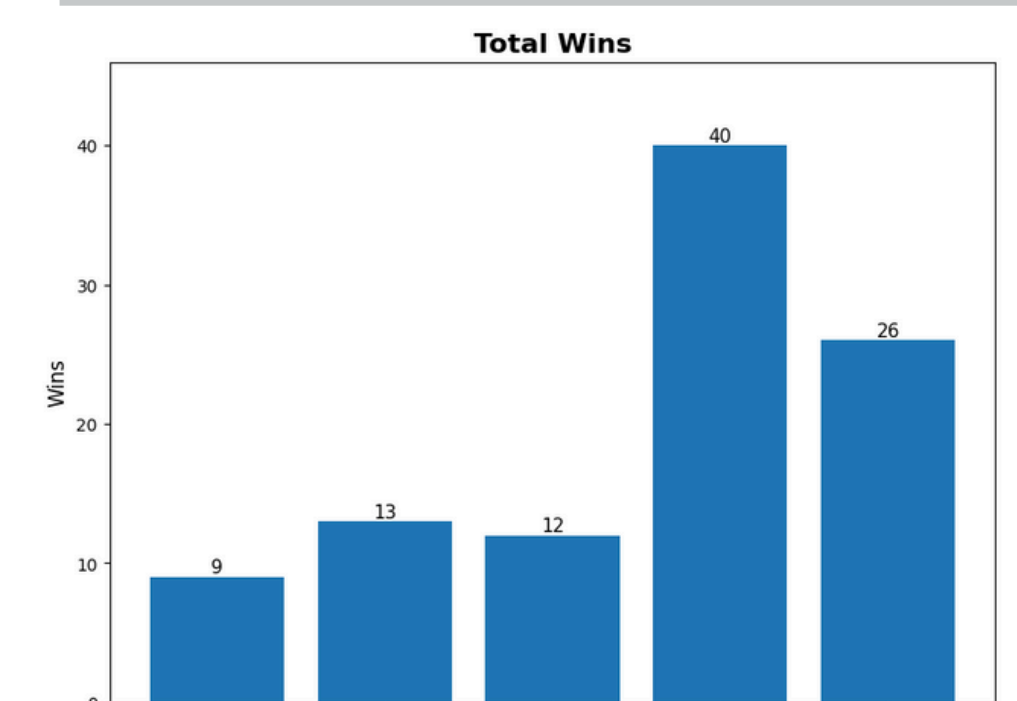
AUDIO

With voice recognition, players can say commands like 'Play C6', which are converted into game actions via Google's API.

Stop Voice Recognition

Listening...

AI BENCHMARK



OUR AI:

Othello AI: Finding the Best Move

Search is performed by simulating move sequences up to depth d , alternating between maximizing and minimizing players.

$$V(s, d, \alpha, \beta, p) = \begin{cases} E(s, p) & \text{if terminal or } d = 0 \\ \max_{a \in A(s, p)} (-V(T(s, a, p), d-1, -\beta, -\alpha, \bar{p})) & \end{cases}$$

s : board state $p, \bar{p}$: player and opponent
 $A(s, p)$: legal moves $T(s, a, p)$: next state after move a

Zobrist Hashing for Fast Caching:

$$h(s) = \bigoplus_{r,c} Z_{r,c}^{(s[r][c])}$$

Compact hashing for each board allows fast lookup in transposition tables.

Dynamic Evaluation Function:

$$\alpha = \frac{\max(0, \min(n - 20, 38))}{38} \quad (\text{game progress})$$

$$W_{r,c} = (1 - \alpha)W_{r,c}^{\text{open}} + \alpha W_{r,c}^{\text{end}} \quad E(s, p) = \sum W_{r,c} + 10 \cdot M + C$$

CORNER STRATEGY:

Corners are **crucial** in Othello as they cannot be flipped once captured, providing stable positions and control over the game

DIFFERENTS AI:

- **RANDOM:** AI that plays randomly
- **MEDIUM:** AI with basic strategy
- **HARD:** AI with advanced strategy
- **OthelloAI:** Our AI
- **NTest:** AI with mobility strategy

MOBILITY:

represents the **number of valid moves available** at each turn. Higher mobility means more strategic options

- Fast search via caching
- Adaptive strategy by phase
- Compact board representation

